R. Brown

39

Tenth Annual In-House Convention

23 & 24 April 1984

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

MONDAY, APRIL 23

1:30 - 3:00 Session I. Chair: R. Klein

1. R. Klein
2. M. Spetch
3. K. Grasse
4. M. Cynader
5. J. McLaren
6. K. Berridge

COFFEE BREAK

3:30 - 5:00 Session II. Chair: ~~Debbie Henken~~ Bill Matheson

7. J. Barresi & E. Wyatt
8. J. Fentress
9. M. Harrington
10. K. Bloom, et al.
11. K. Murphy
12. M. Javilehto

BEER IN LOUNGE

TUESDAY, APRIL 24

9:00 - 10:30 Session III. Chair: Bryan Fantie

13. C. Baker
14. J. Logan & J. Werker
15. K. Briand & L. Smith
16. D. Nicol & I. Meinertzhagen
17. J. Radel
18. S. Shaw & D. Henken

COFFEE BREAK

11:00 - 12:30	Session IV. Chair: Jeff Fairless
	19. L. White & W. Honig 20. J. Matsubara 21. C. Shaw 22. W. Honig & W. Matheson 23. T. Schrans & J. Werker 24. J. Meijer, et al.
12:30 - 2:15	LUNCH & POSTER SESSION
	25. R. Klein, et. al. 26. A. Delamater, et. al. 27. E. Hansen & R. Klein 28. A. Frohlich 40. C. Edwards
2:30 - 3:45	Session V. Chair: ~~Bill Matheson~~ Debbie Harwin
	29. D. Mitchell 30. A. Droungas 31. D Phillips 32. J. Werker & S. Bryson 33. G. James & H. Robertson
3:45 - 4:15	COFFEE BREAK
4:15 - 5:45	Session VI. Chair: R. Rodger
	34. S. Nakajima 35. M. Ozier 36. M. MacLean & S. Bryson 37. J. Kruse 38. R. Brown 39. J. Mendelson
5:30	BEER & DINNER

ABSTRACTS

1. R. Klein

Psychology's In-House Convention: 10 years old and going strong.

The In-House Convention, initiated in 1975 by Graham Goddard, has become an annual departmental event. It provides each of us with an opportunity to present our ideas to a bright and receptive audience and a glimpse at what our departmental colleagues are up to in their labs and minds. A 10 year historical sketch will be presented.

2. M. Spetch

Spatial Memory in Pigeons

why pigeons in a maze?

what is a maple pea?

large value differences?

Previous attempts to demonstrate good spatial working memory in pigeons using the radial arm maze apparatus have met with little success. However, these studies may have inadvertently constrained the pigeon's foraging repertoire. In the present study, the pigeons' task was to locate maple peas which were hidden in eight elevated perches attached to three walls of an 8 x 8 x 10 ft test room. In condition 1, four birds were permitted to choose freely by flying between the perches until all eight peas had been consumed. Choice accuracy improved over trials, was reliably better than chance, and did not appear to depend upon response sequencing. In condition 2, the pigeons could obtain peas from only four randomly selected perches and then were allowed to choose among all eight perches to find the remaining four peas. Again, choice accuracy was above chance. In the last condition a delay of either 3, 30, or 300 sec was imposed after four peas had been obtained. Accuracy in locating the remaining peas declined as a function of delay, but remained above chance. Taken together, three results suggest that pigeons can use spatial memory to forage for food.

3. K. Grasse

Functional Asymmetries in the Oculomotor Reflexes of Frontal and Lateral eyed animals.

When an animal moves around the environment an optical flow field is produced on the retina which provides information concerning the visual consequences of postural adjustments. During locomotion, the predominent vectors (e.g., horizontal or vertical) of self-motion induced optical flow fields depends among other things upon the position of eyes in the head. Some features of this dependency will be discussed.

4. M. Cynader

Functional Topography in Areas 17 and 18 of Cat Visual Cortex

There are a large number of different cortical areas in the cat. We have examined the neuronal response features and topographic organisation of two of these areas, called areas 17 and 18. These areas are different in the way in which they represent the visual field and in receptive field size and prefered velocities of cortical neurons, but bear certain similarities in the way feature extraction properties are laid out across their surfaces.

5. J. McLaren

Hemispheric Asymmetries in the Perception of Emotion as a Function of Task Demands

what do the [illegible]'s indicate

This research investigates hemispheric asymmetries in the perception and identification of both positive and negative emotion. The main purpose was to identify conditions under which an overall left visual field advantage obtains for all emotion and those in which a dissociation is found for positive and negative emotion. Pairs of faces were presented such that one face was presented to each visual field. The pairs consisted of either happy and neutral or sad and neutral faces. Righ-handed subjects were required to respond to the faces that 1) were more emotional, and 2) made them feel either better or worse. The dependent measures were the latency and accuracy of respones. The main finding was that visual field asymmetries depended on the task requirements. When asked to respond to the more emotional face visual field differences varied primarily as a function of the sex of the subject. However, both sexes showed an overall left visual field advantage for better and worse judgements. Support was not provided for the idea of differential hemipsheric specialization for positive and negative emotion. Results are discussed in light of previous findings suggesting that males and females tend to use different strategies in such tasks.

6. K. Berridge

Trigeminal Control Of Ingestive and Grooming Sequences.

fixed action patterns?

nerve regeneration in 3 weeks.

We're currently examining sensorimotor interactions in the production of taste-elicited fixed action patterns and grooming sequences in rats. We're employing a trigeminal deafferentation procedure that eliminates tactile input from the face and mouth while preserving intact all motor efferent and gustatory afferent neurons. This study addresses two issues: 1) the role of non-taste afferent information in the assessment of food palatability, and 2) the role of tactile sensory feedback in the production of ingestive and grooming movements and sequences.

7. J. Barresi & E. Wyatt

Dead or Alive: When Motion Becomes Intention.

perception of meaning in motion, "intentionality" "proto semantics"

A single object moved on a Apple-controlled monitor according to a quantifiable rule. Human observers were asked to evaluate its motion on six Likert Scales: Mechanical, Random, Purposive, Intelligent, Alive, and Human. The proportion of foreward versus 27° turns of the object affected judgements on some of these scales. Whether or not the object occasionally stopped and changed direction also affected judgements. However, complex interactions involving sex of subject and stimulus appearance argue against any simple interpretation of these results.

8. J. Fentress

The Differentiation of Integrated Movement Patterns in Mice and Wolves: A Progress Report.

During development animals (and people) are faced with a dual task. The first challenge is to specify individually isolable dimensions of behaviour from more roughly defined beginnings ("differentiation"). The second challenge is to combine these specified dimensions into coherently ordered patterns of expression ("integration"). In terms of behavioural organization we know surprisingly little of either of these processes, not to mention their relations in time, across levels of operation, etc. Our work on wolves and mice may provide some clues.

9. M. Harrington

Anatomical Studies of the Ventral Lateral Geniculate Nucleus in the Hamster.

The internal dorsal division of the ventral lateral geniculate body (VLGNid) has been shown to contain cells which project to the suprachiasmatic nucleus (SCN) in both rats and hamsters. Cells in the VLGNid show avian pancreatic polypeptide-like (APP-like) immunoreactivity and studies done with rats strongly suggest that the VLGN cells are responsible for most, if not all, of the APP-like immunoreactive fibers observed in the SCN. It has become clear recently that APP-like immunoreactivity observed in the rat brain is probably due to the APP antibody binding to neuropiptide Y (NPY) antigen. I have demonstrated NPY immunoreactive cells in the VLGNid, as well as a dense network of NPY immunoreactive fibers in the SCN of hamsters. I plan to combine wheat germ agglutin-horseradish peroxidase (WGA-HRP) eye injections with NPY immunocytochemistry to demonstrate the relationship of retinal fibers and NPY-immunoreactive VLGN cells. This work is being done in collaboration with Dr. Dwight Nance.

enkephalin?
function?

10. K. Bloom, P. Hazel, A. Russell, H. Walker, & K. Wassenberg

Indirect Effects of Social Reinforcement on the Quality of Infant Vocal Sounds.

Early vocalizations of young humans have the ethological advantage of being amenable to both elicitation and reinforcement by social stimulation (Bloom, 1979). However, social reinforcement, which mimics the properties of dyadic interaction, indirectly modifies the temporal parameters of infant sounds to yield a pattern which parallels "turn-taking" (Bloom, 1977). The results of the present study (28, 3-month-old infants) suggest that social reinforcement also indirectly modifies the quality of vocalizations by increasing the frequency of sounds which adults label as "speech-like".

before 3 mo. of age - the development of mother-infant interaction
- context of expressive language?

11. K. Murphy

<u>Measurement of Vernier Visual Acuity</u>

movement?

Acuity based on the measurement of the minimum perceived misalignment or alignment is termed vernier visual acuity. This is one of the so called hyperacuities because the threshold is only a fraction of the width of the smallest cones in the retina. For humans vernier acuity can be as low as two seconds of arc. Vernier acuity, therefore, is most likely determined cortically and measurement of this type of acuity may demonstrate even more severe visual deficits following visual form deprivation. Recently, Held's group developed a method for testing vernier acuity of human infants which I have modified for measuring kitten's vernier acuity.

12. M. Jarvilehto

<u>A New Techniques to Record and Identify Cells In Tissue Slices From the Insect Visual System.</u>

Anatomical identification of nerve cells combined with their electrical activity forms the basis for the analysis of information processing. The intracellular recording technique has been applied successfully with dye injection procedures on the invertebrate visual ganglia and receptors. Being tedious, uncertain and time-consuming the techniques have now been improved considerably. By using tissue slices of ca. 100 to 300 um, mounted on a special micromanipulator stage, the site of the electrical recording can be viewed under the compound microscope at high power and cells with dye-filling can be identified with great detail by using as a marker the highly fluorescent stain Lucifer Yellow.

13. C. Baker

<u>Temporal Properties of the Short Range Process in Apparent Motion</u>

models of motion detection

Previous studies have examined human perception of random dot two-frame apparent motion while varying the durations of each exposure and the inter-stimulus interval between them. The results are largely consistent with the rule that, for optimal motion detection, a portion of each exposure must fall within the same time interval of about 40 ms. Similar experiments performed on single units in cat area 17 yielded direction-selective responses which behave quite differently, suggesting the possibility that cats perceive motion in a different manner from humans.

14. J. Logan/J. Werker

<u>The Contribution of Acoustic, Phonetic, and Phonemic Factors in Cross-Language Speech Perception.</u>

Speech perception is usually explained by general psychoacoustic or specialized phonetic processes, or as a two-component model with both phonetic and psychoacustic processes. However, our recent cross-language speech perception research has suggested that three factors - acoustic, phonetic, and phonemic - may be necessary to accommodate existing data. A series of experiments performed in our lab were designed to assess this model. English-speaking subjects were tested on their ability to discriminate non-English speech sounds in a same-different discrimination procedure. The interstimulus interval, number of trials, and experimental context were manipulated. Results supported the three-factor model and suggested that the contribution of phonetic processes was secondary to that of acoustic and phonemic processes in accounting for subjects performance.

15. K. Briand & L. Smith

<u>Becker's Verification Model of Word Recognition: A Comparison of Automatic and Controlled Processes.</u>

semantic set? integrative concept?

Becker's verification model of word recognition is an attempt to explain the variability in patterns of facilitation (for related word pairs) and inhibtion (for unrelated word pairs) in lexical decision tasks. The model essentially says that the word recognition, a record of a word held in a temporary sensory store is compared to a number of possible candidates, which are generated either by the visual features of the actual word, or by the semantic features of the context the word is placed in (e.g., previously perceived words). Different patterns of facilitation or inhibition dominance are predicted as a function of the type of semantic context. The present research attempts to demonstrate that Becker's model should not be viewed as a general model of word recognition processes. Rather, it may apply only to processes considered strategic or attention-controlled, and not to those considered to be automatic.

16. D. Nicol & I. Meinertzhagen

<u>Cell Maps and Lineage in the Developing Nervous System of the Ascidian Tadpole Larva.</u>

what happens if one cell is missing?

Larvae of tunicates (Urochordata) have a dorsal tubular nerve cord which is both a symbol of the group's chordate ancestry and a model of the vertebrate nervous system. The nervous system arises from a neural plate, as in vertebrates, but its cellular complement numbers only about 300 and arises from an embryo with strictly mosaic development. The exact lineage of these cells is being followed from detailed cell maps of embryos at different ages and will tell us about some of the factors which generate the fundamental architecture of the vertebrate nervous system. So far the progeny of 10 out of the 12 synchronous divisions have been traced from the egg. Micrographs and reconstructions of various embryonic stages will be presented to illustrate progress.

17. J. Radel

<u>Time course of an Ultrastructural Change and Subsequent Recovery of Normal Morphology in Optic Nerve Fiber Terminals Following Regeneration.</u>

Normal and early regenerate (24-25 days after optic nerve crush, ONC) optic nerve fiber terminals exhibit differences in their patterns of vesicle distribution. These regenerated terminals tend to have a uniform distribution of vesicles throughout the axoplasm, contrasting with the "clustering" of vesicles near retino-tectal synapses in normal optic nerve fiber terminals. This ultrastructural change was shown to also be present in regenerated terminals 4, 6 and 8 months after ONC and to have disappeared by 12 months following ONC. Analysis of the density of vesicles at the retino-tectal synapses and the density of vesicles throughout the terminals revealed that this change was due to a massive influx of vesicles into the axoplasm of the terminals during the first eight months following ONC, rather than to any changes in vesicle density at the retino-tectal synapses. This morphologic change can be used to differentiate original from regenerated optic nerve fiber terminals for up to one year following ONC, particularly in preparations where usual labelling techniques are not feasible.

18. S. Shaw & D. Henken

The Mechanisms Underlying the Blood-Brain Barrier in the CNS of Insects.

Previous work on the visual system had suggested an unusual process of direct electrical interaction between neurones, due to the blockage of current flow across a part of the blood-brain barrier. Curiously, examination of the barrier zone failed to uncover any of the specialized "tight junctions" (TJs) between perineurial cells, that conventional wisdom says form a complete external seal at this type of site in both insects and vertebrates. Puzzled, we therefore re-examined the ventral nerve cord of the CNS, the structure for which the tight junction hypothesis (TJH) was first proposed ca. 1970. Our ultrastructural studies, using serial electronmicroscope sections with a dense extracellular marker (Lanthanum), show directly that the TJH is incorrect. The barrier involved neither perineurial cells nor TJs, and does not appear to form a completely impervious seal. What then forms the barrier? We find that its outer edge coincides with a layer of hitherto undescribed, deeper, sheath-like cells. The measured dimensions of the pathway between these cells predicts a diffusional slowing on a time scale of days: this alone is sufficient to account for the known obstructive properties of the blood-brain barrier.

19. L. White & W. Honig

Category Formation in Baboons

[illegible]

I have trained two female Savannah baboons to choose the odd visual stimulus of the three presented in a modified Wisconsion General Test Apparatus. At present I am using photographs of fruit, leaves, and objects found around the monkey room as visual stimuli. I test for categorization by occasionally giving probe trials in which the three stimuli are different, but two fall within the same category (fruit or leaves). Each baboon shows a tendency to choose the odd member of the trio, thus classifying the two fruits or two leaves together.

[illegible]

20. J. Matsubara

Intrinsic Anatomical Connections Within Visual Cortex and Their Physiological Correlates.

Using neural tracing methods (lectin-HRP, Concanavatin A) we have observed that the local anatomical connections in visual cortex are organized into distinct patterns. The relationship properties and these orderly anatomical connections will be discussed.

21. C. Shaw

Receptors, Development, Plasticity, and All That Jazz.

critical period?

We have devised methods to enable us to selectively locate and study receptors for different neurotransmitters in the visual system. We have examined receptor location in different cortical areas and in different cortical layers within a given area. Some neurotransmitter receptors appear to change their location during the first few months of postnatal life; others do not. The relationship between these findings and the critical period for cortical plasticity will be discussed.

22. W. Honig & W. Matheson

Anticipation of Memory Interval Duration Affects Working Memory

motivation?

Pigeons were trained on two similar working memory tasks that differed in the values of the memory intervals, but had one memory interval in common. The problems were run on different days, with different initial stimuli. Memory performance was strongly affected by the anticipated average duration of the memory interval, so that the birds did much better with the common memory interval from the "favorable" distribution than from the "unfavorable" distribution.

23. T. Schrans & J. Werker

Developmental Influnces on the Awareness of Causal Determinants of Behaviour

sequential integration of social knowledge?

It has been suggested that the deficit in introspective awareness of causal determinants of behaviour noted by Nisbett and Wilson in 1977, may differ in both quality and degree between adults and children. In the present study it was predicted that initially children would demonstrate accurate awareness of the factors influencing their behaviour and this ability would decrease as a function of age and/or cognitive development. The research participants were 31 children; 8 males and 8 females aged 3-4 years, and 7 males and 8 females aged 5-6 years. Each child was required to participate in three experimental tasks over two consecutive days of testing. Task 1 involved standard measures of cognitive development (conservation of number, conservation of area). Task 2 & 3 measured awareness of causal determinants of behaviour (T-shirt experiment, liking judgements). The results supported the hypothesis indicating a developmental trend or inverse relationship between age and introspective awareness of causal determinants of behaviour.

24. J. Meijer, B. Rusak, M. Harrington

synchronization by photic stimuli

Stimulation of a Geniculate Peptidergic Projection Phase Shifts Circadian Rhythms.

The suprachiasmatic nuclei (SCN) function as the dominant pacemaking system in the mammalian circadian system. Direct electrical stimulation what does dark pulses do?

of the SCN mimics the effects of light pulses in causing phase-dependent delays and advances of behavioural rhythms. The ventral nucleus of the lateral geniculate body sends a peptidergic projection to the SCN. Application of this peptide (NPY) to the SCN or electrical stimulation of the lateral geniculate area causes phase-dependnet shifts of behavioural rhythms, but these shifts are different from those caused by light pulses. Instead, they resemble the effects on rhythmicity of dark pulses. These data suggest that release of NPY into the SCN may have the opposite effect of light on SCN neural astinity.

25. R. Klein, D. Pilon, S. Prosser & D. Shannahoff

Nostrils, Hemispheres and Human Performance: Epiligue.

Last year we described EEG data (Werntz et al.) suggesting a link between hemispheric activity and the pattern of air flow through the two nostrils, and we presented the design of an experiment to determine if human performance on hemisphere specific (verbal and spatial) tasks would be appropriately influenced by which nostril is breathing. Two such experiments, testing over 120 subjects, have now been completed. The relative performance on the verbal and spatial tasks was (weakly) correlated with the pattern of nasal congestion (the nasal cycle), but 30 minutes of uninostril breathing had little effect on performance. POSTER

26. A. Delamater, J. Kruse & V. LoLordo

Pavlovian Conditoned Inhibition In Taste Aversion Learning Revisited

Several assays were used in assessing conditioned inhibition within a taste aversion procedure. Following A+, AX- training, X (NaCl) was not consistently preferred to water or diluted quinine in two-bottle tests, nor did retardation or summation tests yield consistent outcomes. We question the usefulness of CTAs for studying Pavlovian conditoned inhibition. POSTER

27. E. Hansen & R. Klein

Dual Task Analysis of Cognitive Control of Visual Attention.

A priming paradigm was used to induce attentional shifts towards likely locations of imperative visual stimuli. Manual or saccadic responses systems showed equal costs and benefits in a single task design. Infrequent responses did not show costs and benefits in a dual task situation. A factorial set of experiments manipulated whether infrequent stimuli had to be discriminated from frequent ones, and whether primes were informative for infrequent stimuli. Costs and benefits for both stimuli were equal in the single response experiments; in the dual response experiments, infrequent stimuli/responses did not show costs and benefits. The assumption that locational cues orient attention towards simply a locus in space is incomplete. POSTER

28. A. Frohlich

Freeze-Fracture of Visual Receptor Synapses in the Fly.

Photoreceptor axon terminals in the lamina ganglionaris, the first optic neuropile of the housefly Musca domestica, were examined using freeze-fracture technique. The p-face of the terminal membrane contains numerous bow-tie shaped particle clusters representing the presynaptic site. In some fractures the remains of postsynaptic elements can be seen opposite these sites displaying several characteristic features, e.g., presumptive intramembraneous transmitter receptor particles. POSTER

29. D. Mitchell

Comparison of the Effects On Visual Acuity of Ablation of Visual Cortical Areas 17 and 18 in Kittens or Cats of Different Ages.

Previus work has shown that following complete excision of cortical area 17 and 18 in adult cats visual acuity is reduced by about a factor of two. However, comparable lesions made in infancy at between 10 and 27 days of age produce little or no deficits in visual acuity. Surprisingly, I have recently found that lesions made even earlier, on the third day of postnatal life, produce deficits comparable to those observed in animals lesioned in adulthood. This paradoxical finding may be related to the loss of X type retinal ganglion cells reported by others in animals lesioned on or about the day of birth.

30. A. Droungas

Possible involvement of Opponent Processes in the Development of Inhibition in Backward Conditioning.

Backward conditioning (US - CS interval 0.5 sec) was found to be more effective than explicitly unpaired training (US - CS interval 7 min) in resulting in inhibition when the inter-trial interval was fixed and equal in the two procedures (ITI = 7 min). Inhibition in backward conditioning was augmented when the ITI was varied unpredictably across trials; inhibition was restored when US pre-exposure preceded backward conditioning with the variable ITI. These results are discussed in view of the opponent process theory of acquired motivation (Solomon and Corbit, 1974).

31. D. Phillips

A Neural Mechanism in the Cat's Auditory Cortex Underlying Sensitivity to Direction of Frequency Change.

Single neurones in the cat's primary auditory cortex have restricted frequency 'response areas' within which a tone of constant frequency is able to elicit spike discharges. We have examined the sensitivity of these cells to ongoing tones in which we embedded a small (2 kHz) frequency-modulated (FM) ramp. The absolute frequency range traversed by the FM was varied in and around each cell's response area. Irrespective of their other properties, cortical neurones responded to these FM ramps with a profound direction selectivity. These neurones responded to FM ramps only if direction of frequency change was towards the centre of their response areas. The strength of directional preference depended on the amount of the response area covered by the FM ramp, and on the rate of change of the ramp's instantaneous frequency's ability to elicit spike discharges.

32. J. Werker & S. Bryson

Toward an Understanding of Consonant Errors in Reading and Spelling: Severely Disabled and Normal Readers.

This research examined how severely disabled readers read and spell common English consonants. Three groups of disabled readers were compared to both age and reading level-matched control groups. Consonant errors were analyzed according to five error types. Results replicated previous findings showing that most consonant errors can be accounted for by phonetic rather than visual substitutions. Moreover, all three groups of severely disabled readers made as many addition errors as they did phonetic substitutions, and significantly more addition errors than did the normal readers. The pattern of addition errors suggests the use of an articulatory strategy in reading.

33. G. James & H. Robertson

An Animal Model for Schizophrenia: Effects of Chronic Phencyclidine Treatments on Levels of Dopamine and DOPAC in the Extrapyrimidal and Mesolimbic Systems of the Rat.

Chronic treatment with phencyclidine for three weeks caused changes in the levels of dopamine and DOPAC in the striatal and mesolimbic systems. The nature and extent of the change was related to the length of time the sujects were drug free prior to sacrifice. The levels of dopamine were increased shortly after phencyclidine treatment, but then drop to normal after 51 days drug free. Similarly, the levels of POPAC also fall after 21 days drug free to below control levels. Haloperidol challenge 1 hour prior to sacrifice was found to have little effect on the levels of either dopamine or DOPAC in both striatum and nucleus accembens.

34. S. Nakajima K O'Connor, H. Robertson

Dopamine Receptors Involved in Self-Stimulation.

D1 - cAMP
D2 - no cAMP

Self-stimulation is suppressed by various substances that block dopamine receptors. We now report that a drug (SCH23390) specific to D1 receptors suppresed self-stimulation in the rat with hypothalamic electrode but another drug (sulpiride) specific to D2 receptor did not. (We are grateful to Dr. G. McKenzie for SCH233390 and to Mr. DeLaGrange for sulpiride) dry spouts -

35. M. Ozier

Induced Depression and Elation and the Episodic Memory Trace

Induced depression and elation may have differential effects upon automatic and effortful encoding of episodic memory codes. An experiment to test this hypothesis was performed and the data provide strong support for the notion that there are two modes of encoding episodic information.

36. M. MacLean & S. Bryson   i.e. from results?

<u>Hemispheric Differences in the Perception of Emotional Valence.</u>

This study investigated hemispheric differences in the perception of emotional valence. Right-handed, male and female subjects rated the degree of happy or sad emotion displayed in a series of happy, sad and neutral expressions. The faces were presented laterally and in the central field of vision for a duration of 60, 120 and 180 msecs. Independent of both the exposure duration and the emotional expression, faces presented to the left visual field were rated more negatively than those projected to the right visual field. Moreover, the left visual field ratings for the females were even more negative than those for the males. No visual field differences were found in the ability to distinguish among emotional expressions. The findings are discussed in terms of the idea that the two cerebral hemisheres differ in their emotional responses.

37. J. Kruse

<u>Pavlovian Factors in Thirst.</u>

Rats initially prefer .9% (isotonic) saline to water, but protracted experience with saline as the sole fluid reverses this preference. The present experiment extended the Pavlovian analysis of this phenomenon by including an overshadowing group, which experienced salt mixed with vanilla as their sole fluid between the two-bottle pretest and posttest for salt preference. As expected, this group showed an attenuated saline aversion relative to a group which drank pure saline solution duirng the training interval. In addition, a vanilla aversion developed in the former group. The results suggest that thirst can serve as a US in Pavlovian preparations.

38. R. Brown

<u>Preferences for Maternal Odours: 500 Mice Can't Be Wrong.</u>

Olfactory signals have the potential to contain many different types of information. They can be used to identify the species, sex, age and even the individuality of an animal as well as its social status, breeding condition, stress level and maternal state. "Maternal odours" excreted in the feces of lactating female rats and mice may contain information about sex and individuality and can be modified by maternal diet. Five experiments were done to examine the influence of diet on preferences of 18-20 day old CD-1GF albino mice for maternal odours, odours of non-lactating females, males, and strange mothers. The results suggest that infant mice develop graded preferences for different components of the "maternal odour". This supports the hypothesis that maternal diet may increase the preferences of pups for other adults on the same diet.

39. J. Mendelson

Selectivity of Cells in Cat Superior Colliculus for Features of Simulated Sound Movement.

The superior colliculus (SC) is a multimodal structure which appears to be involved in orientation toward objects in the environment. The superficial layers processd visual information only while the intermediate and deep layers process a combination of visual, auditory, and somatosensory information. Most research in the area has been devoted to studying the SC's responses to visual stimuli; comparatively little has been devoted to the processing of stimuli from the other two modalities. We have recently embarked on a series of experiments which will study the effects of simulated sound movement on cells in the deep layers of the SC. Preliminary data reveal that the collicular responses are remarkably similar to those of the auditory cortex. These data re relevant to investigations of the interaction between auditory and visual stimuli on collicular unit responses.

40. C. Edwards (POSTER)